

TOP 15 LEETCODE INTERVIEW QUESTIONS

Java Placement Preparation

Array | String | Linked List | Stack | DP | Graph | Heap | Design



Scan to Verify / Connect

Dr. Amar Panchal

Technical Trainer & Founder, Career Credentials

 9821601163

 amar@careercredentials.in



Q1. Two Sum

Difficulty: **Easy** | Category: **Array / HashMap**

Problem Statement:

Given an array of integers `nums` and an integer `target`, return indices of the two numbers such that they add up to `target`. You may assume that each input would have exactly one solution, and you may not use the same element twice.

Example:

```
Input: nums = [2,7,11,15], target = 9
Output: [0,1]
Explanation: nums[0] + nums[1] = 2 + 7 = 9
```

Constraints:

- $2 \leq \text{nums.length} \leq 10^4$
- $-10^9 \leq \text{nums}[i] \leq 10^9$
- Only one valid answer exists

✍️ *Write Your Solution Below:*

Q2. Valid Parentheses

Difficulty: **Easy** | Category: **Stack**

Problem Statement:

Given a string `s` containing just the characters '(', ')', '{', '}', '[' and ']', determine if the input string is valid. An input string is valid if: open brackets are closed by the same type, open brackets are closed in the correct order, and every close bracket has a corresponding open bracket.

Example:

```
Input: s = "()[]{}" → Output: true
Input: s = "]" → Output: false
```

Constraints:

- $1 \leq \text{s.length} \leq 10^4$
- `s` consists of parentheses only '()[]{}'

✍️ *Write Your Solution Below:*



Scan to Verify / Connect

Dr. Amar Panchal

Technical Trainer & Founder, Career Credentials

📞 9821601163

✉️ amar@careercredentials.in



Q3. Best Time to Buy and Sell Stock

Difficulty: **Easy** | Category: **Array / Greedy**

Problem Statement:

You are given an array prices where prices[i] is the price of a given stock on the i-th day. You want to maximize your profit by choosing a single day to buy one stock and choosing a different day in the future to sell that stock. Return the maximum profit you can achieve. If you cannot achieve any profit, return 0.

Example:

Input: prices = [7,1,5,3,6,4]

Output: 5

Explanation: Buy on day 2 (price=1), sell on day 5 (price=6). Profit = 6-1 = 5

Constraints:

- $1 \leq \text{prices.length} \leq 10^5$
- $0 \leq \text{prices}[i] \leq 10^4$

— Write Your Solution Below:

Q4. Merge Two Sorted Lists

Difficulty: **Easy** | Category: **Linked List**

Problem Statement:

You are given the heads of two sorted linked lists list1 and list2. Merge the two lists into one sorted list. The list should be made by splicing together the nodes of the first two lists. Return the head of the merged linked list.

Example:

Input: list1 = [1,2,4], list2 = [1,3,4]

Output: [1,1,2,3,4,4]

Constraints:

- Number of nodes in both lists is in range [0, 50]
- $-100 \leq \text{Node.val} \leq 100$
- Both lists are sorted in non-decreasing order

— Write Your Solution Below:



Scan to Verify / Connect

Dr. Amar Panchal

Technical Trainer & Founder, Career Credentials

9821601163

amar@careercredentials.in



Q5. Maximum Subarray (Kadane's Algorithm)Difficulty: **Medium** | Category: **Array / DP****Problem Statement:**

Given an integer array `nums`, find the subarray with the largest sum, and return its sum. A subarray is a contiguous non-empty sequence of elements within an array.

Example:Input: `nums = [-2,1,-3,4,-1,2,1,-5,4]`

Output: 6

Explanation: The subarray `[4,-1,2,1]` has the largest sum = 6**Constraints:**

- `1 <= nums.length <= 10^5`
- `-10^4 <= nums[i] <= 10^4`

Write Your Solution Below:**Q6. Reverse a Linked List**Difficulty: **Easy** | Category: **Linked List****Problem Statement:**

Given the head of a singly linked list, reverse the list, and return the reversed list.

Example:Input: `head = [1,2,3,4,5]`Output: `[5,4,3,2,1]`**Constraints:**

- Number of nodes in the list is in range `[0, 5000]`
- `-5000 <= Node.val <= 5000`

Write Your Solution Below:

Scan to Verify / Connect

Dr. Amar Panchal

Technical Trainer & Founder, Career Credentials

 9821601163 amar@careercredentials.in

Q7. Climbing Stairs

Difficulty: **Easy** | Category: **Dynamic Programming**

Problem Statement:

You are climbing a staircase. It takes n steps to reach the top. Each time you can either climb 1 or 2 steps. In how many distinct ways can you climb to the top?

Example:

```
Input: n = 3
Output: 3
Explanation: 1+1+1, 1+2, 2+1
```

Constraints:

- $1 \leq n \leq 45$

— Write Your Solution Below:

Q8. Binary Search

Difficulty: **Easy** | Category: **Binary Search**

Problem Statement:

Given an array of integers `nums` which is sorted in ascending order, and an integer `target`, write a function to search `target` in `nums`. If `target` exists, then return its index. Otherwise, return -1. You must write an algorithm with $O(\log n)$ runtime complexity.

Example:

```
Input: nums = [-1,0,3,5,9,12], target = 9
Output: 4
```

Constraints:

- $1 \leq \text{nums.length} \leq 10^4$
- All elements in `nums` are unique
- `nums` is sorted in ascending order

— Write Your Solution Below:



Scan to Verify / Connect

Dr. Amar Panchal

Technical Trainer & Founder, Career Credentials

9821601163

amar@careercredentials.in



Q9. Number of IslandsDifficulty: **Medium** | Category: **Graph / DFS / BFS****Problem Statement:**

Given an $m \times n$ 2D binary grid which represents a map of '1's (land) and '0's (water), return the number of islands. An island is surrounded by water and is formed by connecting adjacent lands horizontally or vertically.

Example:

```
Input:
11110
11010
11000
00000
Output: 1
```

Constraints:

- $m == \text{grid.length}$, $n == \text{grid}[i].\text{length}$
- $1 \leq m$, $n \leq 300$
- $\text{grid}[i][j]$ is '0' or '1'

➡ **Write Your Solution Below:**

Q10. Longest Common PrefixDifficulty: **Easy** | Category: **String****Problem Statement:**

Write a function to find the longest common prefix string amongst an array of strings. If there is no common prefix, return an empty string "".

Example:

```
Input: strs = ["flower", "flow", "flight"]
Output: "fl"
```

Constraints:

- $1 \leq \text{strs.length} \leq 200$
- $0 \leq \text{strs}[i].\text{length} \leq 200$
- $\text{strs}[i]$ consists of only lowercase English letters

➡ **Write Your Solution Below:**



Scan to Verify / Connect

Dr. Amar Panchal

Technical Trainer & Founder, Career Credentials

 9821601163 amar@careercredentials.in

Q11. 3Sum

Difficulty: **Medium** | Category: **Array / Two Pointers**

Problem Statement:

Given an integer array `nums`, return all the triplets `[nums[i], nums[j], nums[k]]` such that $i \neq j$, $i \neq k$, $j \neq k$, and $nums[i] + nums[j] + nums[k] == 0$. The solution set must not contain duplicate triplets.

Example:

```
Input: nums = [-1,0,1,2,-1,-4]
Output: [[-1,-1,2],[-1,0,1]]
```

Constraints:

- $3 \leq \text{nums.length} \leq 3000$
- $-10^5 \leq \text{nums}[i] \leq 10^5$

— Write Your Solution Below:

Q12. LRU Cache

Difficulty: **Medium** | Category: **Design / LinkedHashMap**

Problem Statement:

Design a data structure that follows the constraints of a Least Recently Used (LRU) cache. Implement the `LRUCache` class with `get(key)` and `put(key, value)` methods. Both must run in $O(1)$ average time complexity. `get` returns the value or `-1` if not found. `put` inserts or updates the key-value pair, evicting the least recently used key when capacity is exceeded.

Example:

```
LRUCache lru = new LRUCache(2);
lru.put(1,1); lru.put(2,2); lru.get(1) → 1;
lru.put(3,3); lru.get(2) → -1 (evicted)
```

Constraints:

- $1 \leq \text{capacity} \leq 3000$
- $0 \leq \text{key} \leq 10^4$
- Up to 2×10^5 calls to `get` and `put`

— Write Your Solution Below:



Scan to Verify / Connect

Dr. Amar Panchal

Technical Trainer & Founder, Career Credentials

9821601163

amar@careercredentials.in



Q13. Word Search

Difficulty: **Medium** | Category: **Backtracking / DFS**

Problem Statement:

Given an $m \times n$ grid of characters `board` and a string `word`, return true if `word` exists in the grid. The word can be constructed from letters of sequentially adjacent cells, where adjacent cells are horizontally or vertically neighboring. The same letter cell may not be used more than once.

Example:

```
Input: board = [["A","B","C","E"],["S","F","C","S"],["A","D","E","E"]], word = "ABCCED"
Output: true
```

Constraints:

- $m == \text{board.length}$, $n == \text{board}[i].\text{length}$
- $1 \leq m, n \leq 6$
- `word.length` between 1 and 15

✍️ *Write Your Solution Below:*

Q14. Find Median from Data Stream

Difficulty: **Hard** | Category: **Heap / Priority Queue**

Problem Statement:

The `MedianFinder` class is used to find the median of a stream of integers. Implement `addNum(int num)` to add a number to the data structure, and `findMedian()` to return the median. If the size of the list is even, the median is the mean of the two middle values.

Example:

```
addNum(1) → addNum(2) → findMedian() = 1.5
addNum(3) → findMedian() = 2.0
```

Constraints:

- $-10^5 \leq \text{num} \leq 10^5$
- At least one element added before `findMedian()`
- At most $5 \cdot 10^4$ calls to `addNum` and `findMedian`

✍️ *Write Your Solution Below:*



Scan to Verify / Connect

Dr. Amar Panchal

Technical Trainer & Founder, Career Credentials

📞 9821601163

✉️ amar@careercredentials.in



Q15. Trapping Rain Water

Difficulty: **Hard** | Category: **Array / Two Pointers / Stack**

Problem Statement:

Given n non-negative integers representing an elevation map where the width of each bar is 1, compute how much water it can trap after raining.

Example:

Input: `height = [0,1,0,2,1,0,1,3,2,1,2,1]`

Output: 6

Explanation: The diagram shows 6 units of trapped rain water.

Constraints:

- $n == \text{height.length}$
- $1 \leq n \leq 2 \cdot 10^4$
- $0 \leq \text{height}[i] \leq 10^5$

➡ **Write Your Solution Below:**



Scan to Verify / Connect

Dr. Amar Panchal

Technical Trainer & Founder, Career Credentials

📞 9821601163

✉️ amar@careercredentials.in

